

Distributed Systems

COMP 212

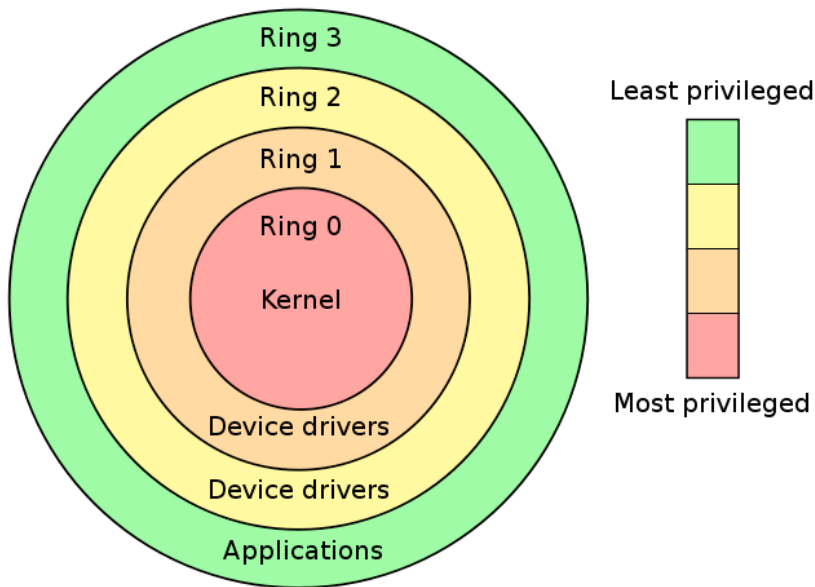
Lecture 23

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Virtualisation & Cloud Computing

Protection rings

- It's all about **protection rings** in modern processors
- Hardware mechanism to protect data and functionality from faults and malicious behaviour



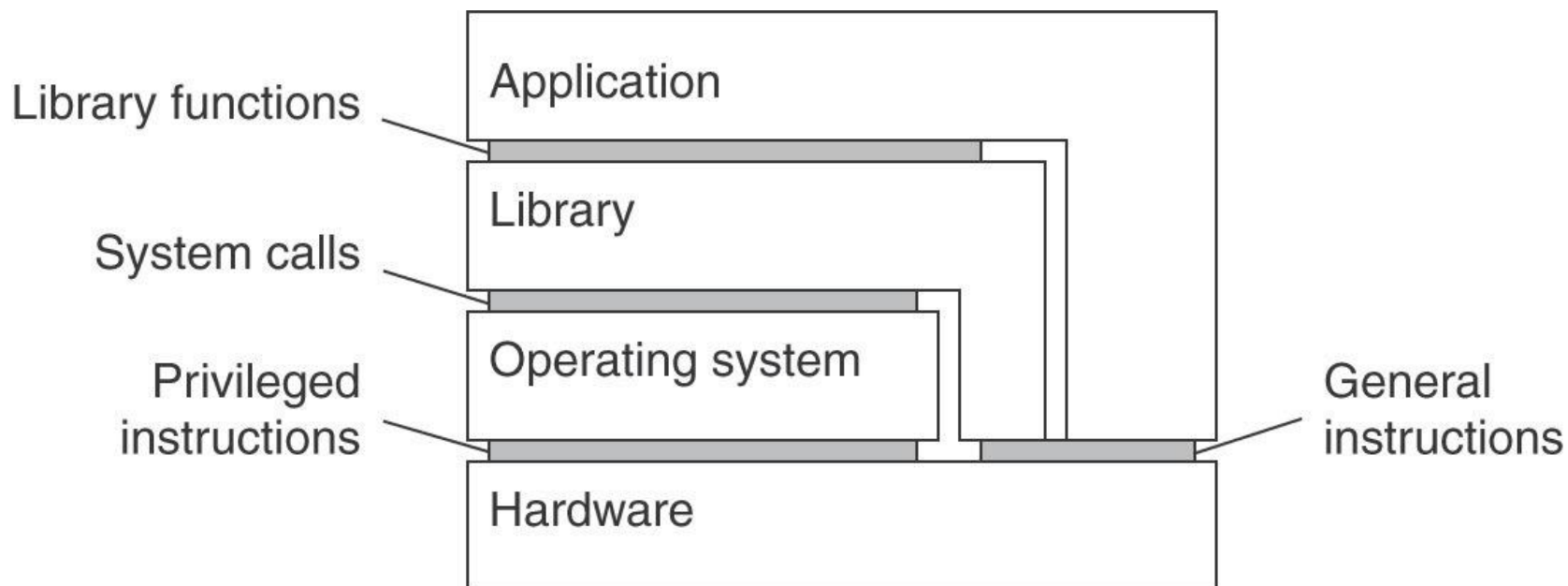
- x86 processor can function in one of 4 modes
- Modern operating systems only use Ring 0 and Ring 3

x86 protection rings

http://upload.wikimedia.org/wikipedia/en/2/2f/Priv_rings.svg

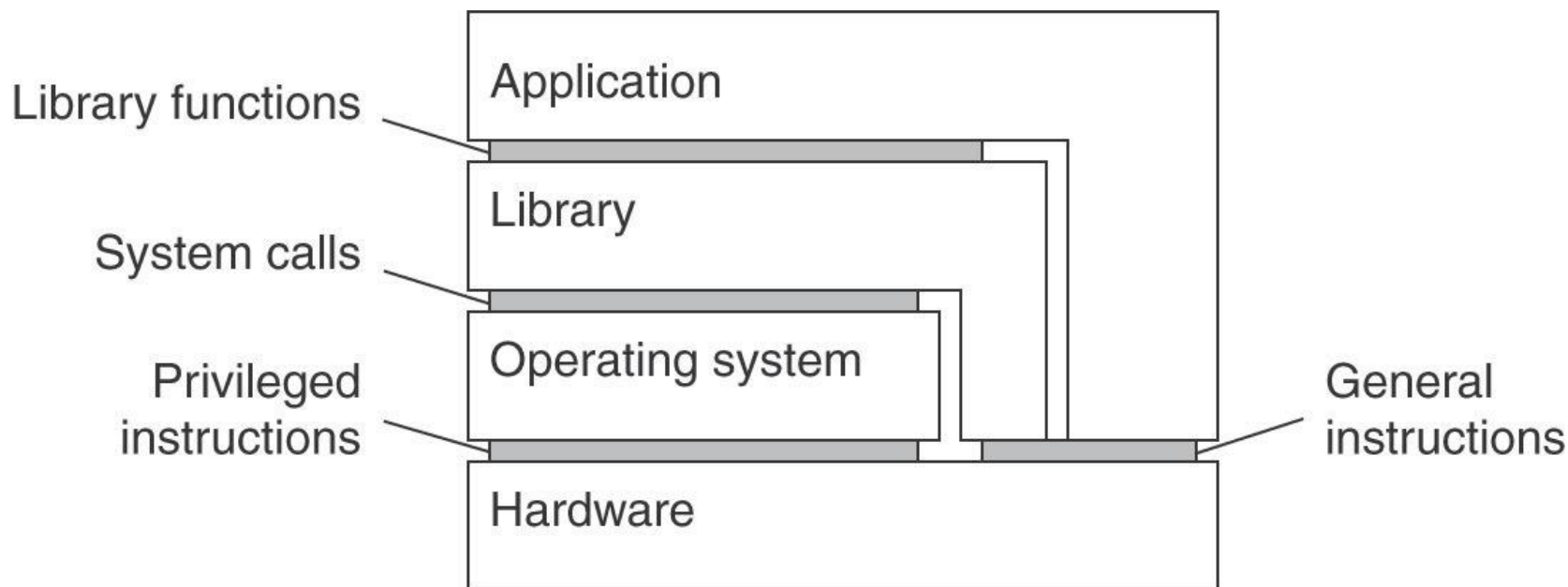
Interfaces at different levels (1)

- Hardware ↔ software **general** machine instructions
 - that can be invoked by any program.
- Hardware ↔ software **privileged** machine instructions
 - that can be invoked only by privileged programs, such as an operating system.



Interfaces at different levels (2)

- **System calls** offered by an operating system
 - A programmatic way in which a program requests a service from the kernel of the OS it is executed on (e.g. accessing a disk drive).
- **Library calls**
 - generally forming what is known as an application programming interface (API)

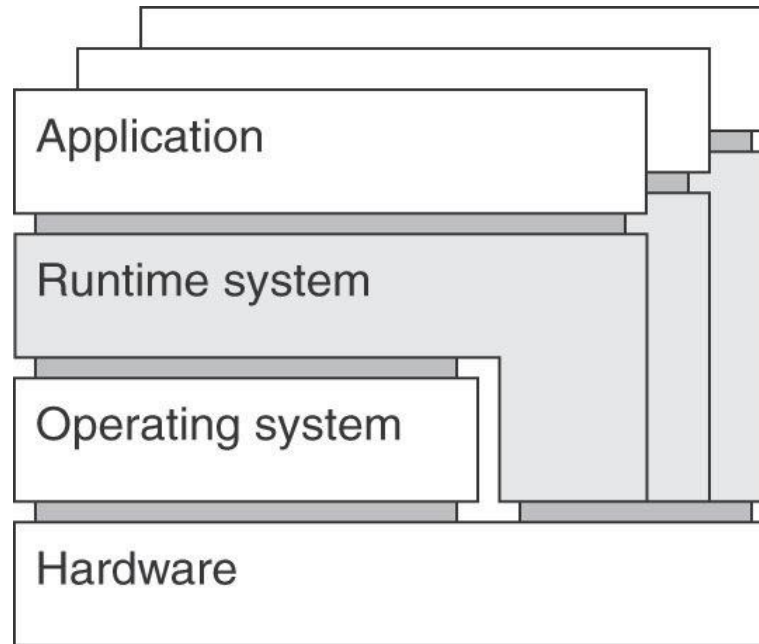


Virtualization in a Nutshell

- Implementation of a virtual, instead of a physical version of a server, a storage device, an OS, etc.
- Threads and (mainly) processes maintain a virtual environment for a task (context)
 - An illusion of parallelism is created
 - Resource virtualization
- Other examples of virtualization:
 - Storage virtualization (your CS network disk)
 - Virtual memory (processes have own address space)
 - Virtual machines (e.g. Java)
 - VM Ware
- Virtualisation deals with extending or replacing an existing interface so as to mimic the behaviour of another system.

Architectures of Virtual Machines (1)

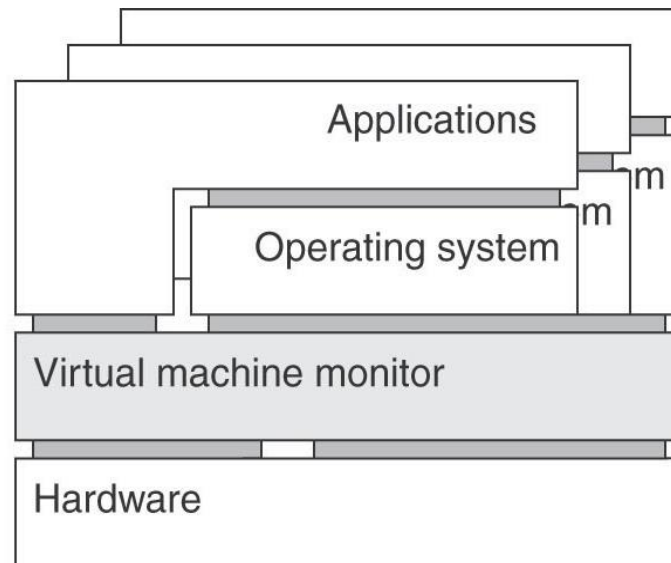
- Process Virtualisation (single process)
 - Build a runtime system that provides an abstract instruction set.



(a)

Architectures of Virtual Machines (2)

- System Virtualisation (multiple processes)
 - Provide a system that is implemented as a layer, completely shielding the original hardware.
 - Offering the complete instruction set of the same (or another) hardware as an interface.
 - Simultaneous use of this interface by different programs (multiple OS run independently on the same platform)



(b)

Desktop Virtualisation Benefits

- Desktop virtualisation, typically, allows one to run an entire (guest) operating system as a process within the (host) operating system controlling the hardware
- E.g. Ubuntu Linux in VirtualBox running under Windows
- Access to new and experimental technology
- Easy network programming
- Portability checks
 - E.g. checking that your JavaScript works in IE6, IE7,.. Mozilla, Firefox,...

Server Virtualisation Benefits

- Multiple servers live on a single physical machine.
- Abstraction
 - Hide physical characteristics of hardware
- Isolation
 - Run several “logical” servers on a single physical server
 - Easily create heterogeneous environment
- Replication
- Reliability and scalability

Server Virtualisation Scenario (1)

- Consider a multi-tier heterogeneous system
- Requires three different machines to run
 - Prone to failure
 - Poor maintainability



Microsoft
IIS web
server



Linux
application
server



Oracle DB

Server Virtualisation Scenario (2)

- Consider a **virtualised** multi-tier heterogeneous system
- Can run on one computer
 - Prone to failure
- Can spread over a **computer cluster**
 - Fault tolerant
 - Scalable
 - More stable



Microsoft
IIS web
server



Linux
application
server

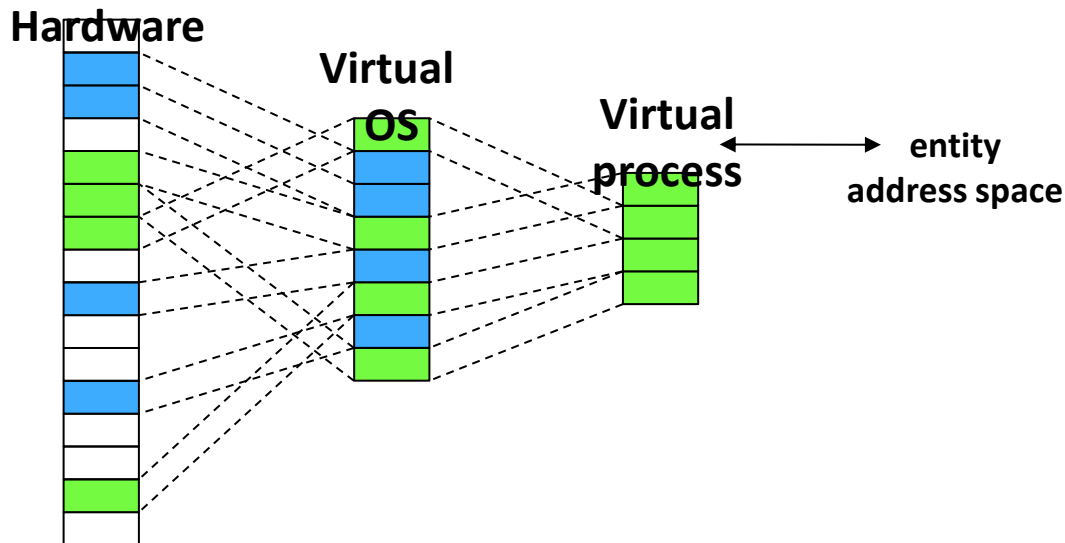


Oracle DB

Hardware-Assisted VMM

Hardware-assisted virtualisation benefits from

- Running general instructions directly on the hardware
- Trapping privileged instructions



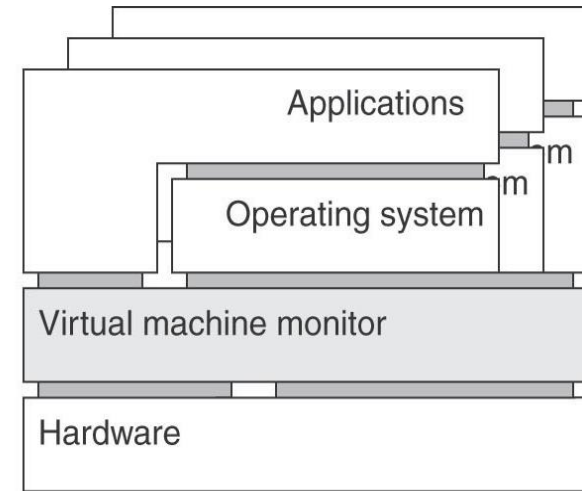
VMM Principles

Popek and Goldberg (1974):

- Equivalence
 - Guest system should run in the same way as if the environment was not virtualised
- Resource control
 - VMM takes full control over the virtualised resources
- Efficiency
 - Majority of machine instructions should be executed without VMM interventions

Virtual Machine Monitor

- VMM intercepts operations that interfere with the host hardware
- Higher level of control than the operating system
 - Hardware support (Ring -1)
 - Intel VT-x, AMD-V
- Software virtualisation
 - Modify guest OS
 - Xen hypervisor
 - Software emulation (slow)
 - Modifying (parts of) guest code on the fly



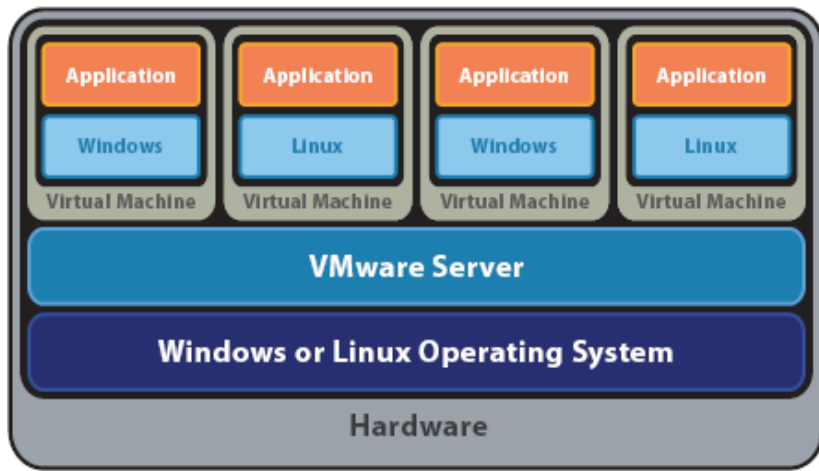
(b)

System Snapshot

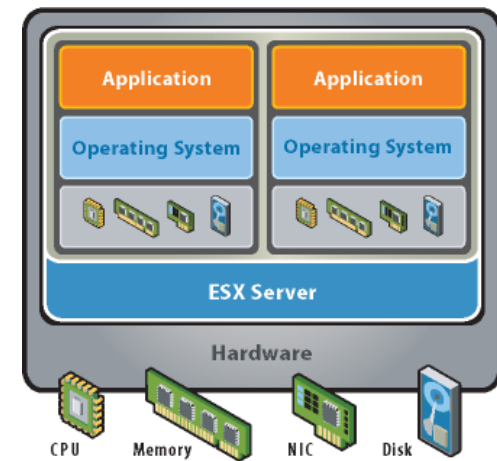
- A VMM can Stop/Freeze/Resume a VM
- Take a snapshot, archive and rollback
- Move/replicate a VM
 - Downloadable “appliances”
- Live Migration
 - Until moved
 - Stop the source machine
 - Copy some information
 - Resume the machine
- Increased availability & load balancing

Example: VMWare

- Workstation / Fusion / Player
- VMWare Server (GSX server) runs over a Linux/Windows host
- VMWare Server (ESX server) runs natively

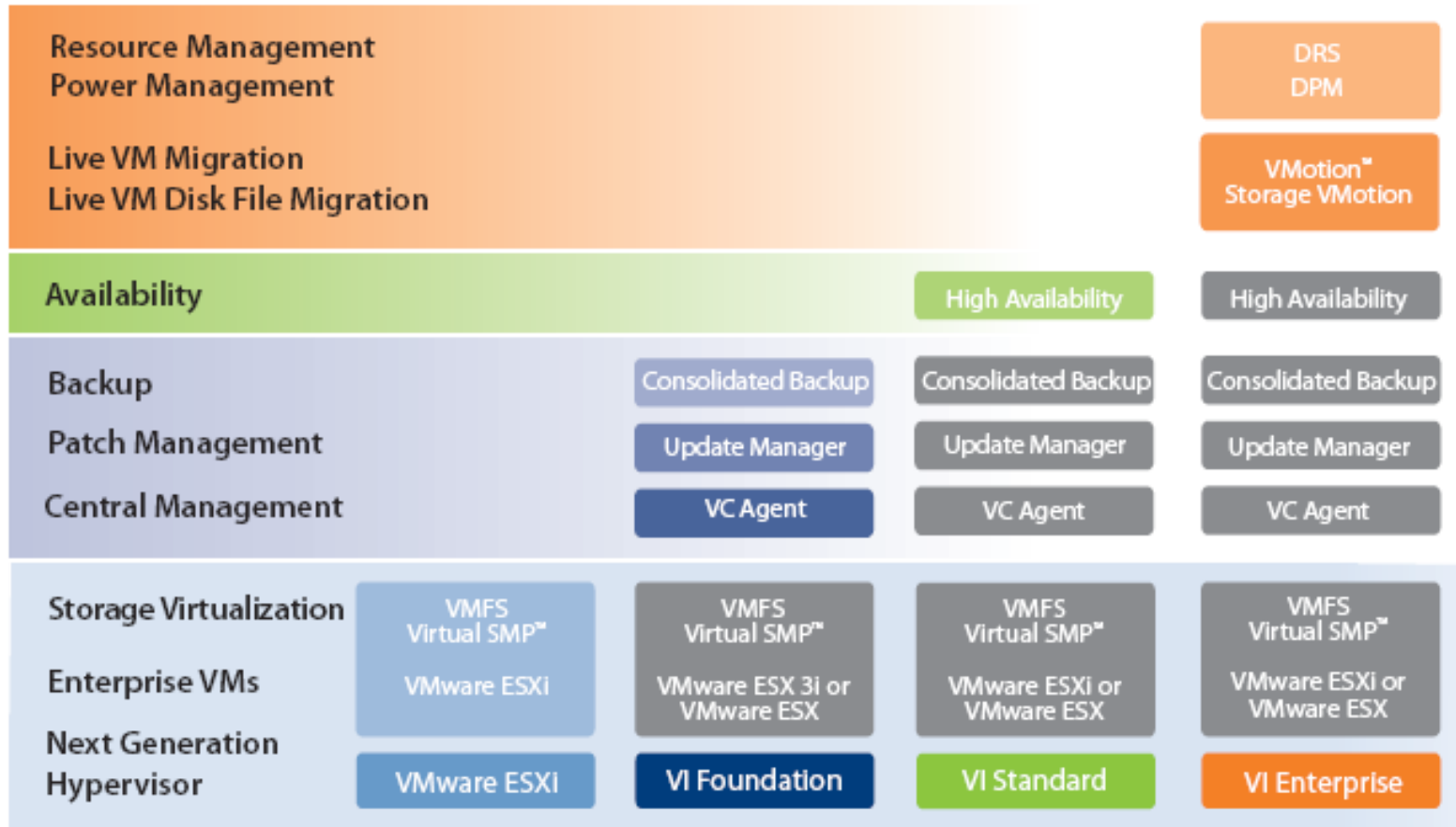


Hosted virtualisation



Bare-metal virtualisation

VMWare Infrastructure



Example: Microsoft Hyper-V

- Bare-metal solution
- Requires a copy of Windows Server in one of partitions
- Supports Windows and Linux guests
 - Microsoft produced a Linux driver to improve performance

Example: VirtualBox

- Mainly desktop virtualization solution
- The bulk of it is free software
 - Contains proprietary parts
- Software virtualization
 - Runs the guest OS in Ring 1 (not used otherwise by many systems)
 - Requires fiddling on the fly
- Hardware-assisted virtualization
 - Runs a Ring -1 VMM
 - Guest OS runs in Ring 0 (no modification needed)

Example: Xen

Runs on bare metal

- Must have a “master” copy of OS (Linux or FreeBSD)
- Guest OS's are aware of being run in the virtualised environment – **paravirtualisation**
- Offers both native and simulated hardware interfaces
- Now support hardware-assisted virtualisation

Unknown Resource Demands

- **Scenario:** a start-up company requires computing facilities
- Number of users — ??
- Load on servers — ??
- IT budget — ??

Answer: On-demand resource provisioning

Cloud Computing

Virtual machines in the cloud

- Resizable computing capabilities
- Resizable storage
- You get a virtual machine to use it as you like
 - (subject to terms and conditions)
- Service provider maintains the infrastructure
- You pay for what you consume

There are other forms of cloud computing

- Google apps, ...

Example: Amazon EC2

Amazon Elastic Compute Cloud:

- Started as Amazon's own effort to service its customers
- Amazon maintains and supports the infrastructure
 - Uses Xen virtualization
- Coupled with Amazon Simple Storage Service (Amazon S3)
- There are other cloud solutions

Note

Remember that

- Moving your service into the cloud alone will not achieve scalability.
- The cloud provides **means** for
 - Size scalability
 - Geographical scalability
- You still need to manage scalability

Virtualisation & CC: Advantages

- Abstraction, isolation, distributed transparency
- Reliability, scalability (elasticity)
- Easy to create new “clean” machines
- Run legacy software
- Test new setups, configurations, updates,...
- Lower computer costs

Virtualisation & CC: Disadvantages

- System can work differently in virtual environment
- Might not reach peak performance
- Security and privacy
- Loss of control
- Requires constant high-speed Internet
- Hard to move back (+loss of expertise)